Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the accuracy during training.

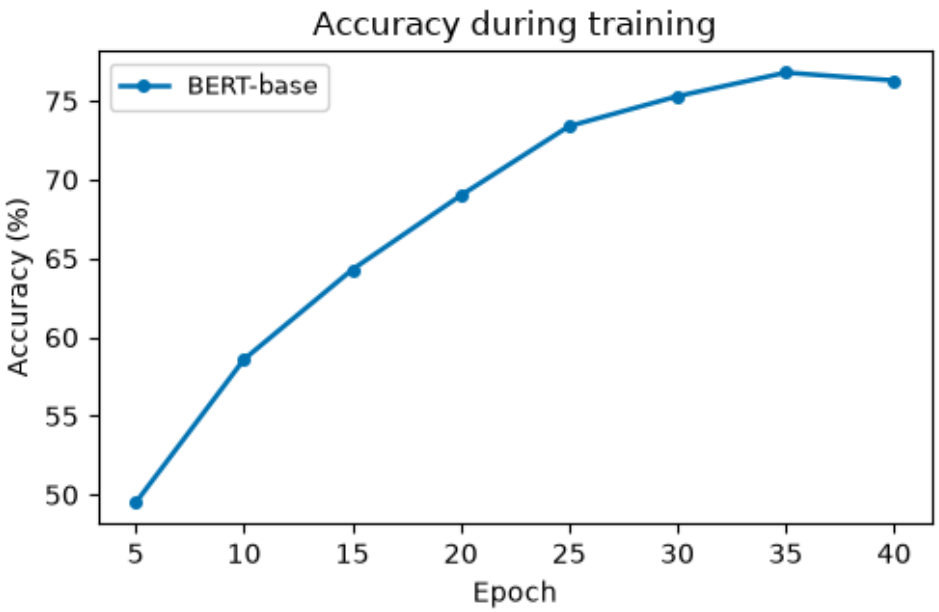


Figure 1: Accuracy during training for the models in this experiment.

As shown in Figure 1, BERT-base scores 10.6 points higher on 10 than on 5. BERT-base scores about 3 points higher on 35 than on 25. At epoch 30, BERT-base achieves 75.9% accuracy. At epoch 15, BERT-base achieves 70.3% accuracy.

Figure 2 shows the f1 score during training.

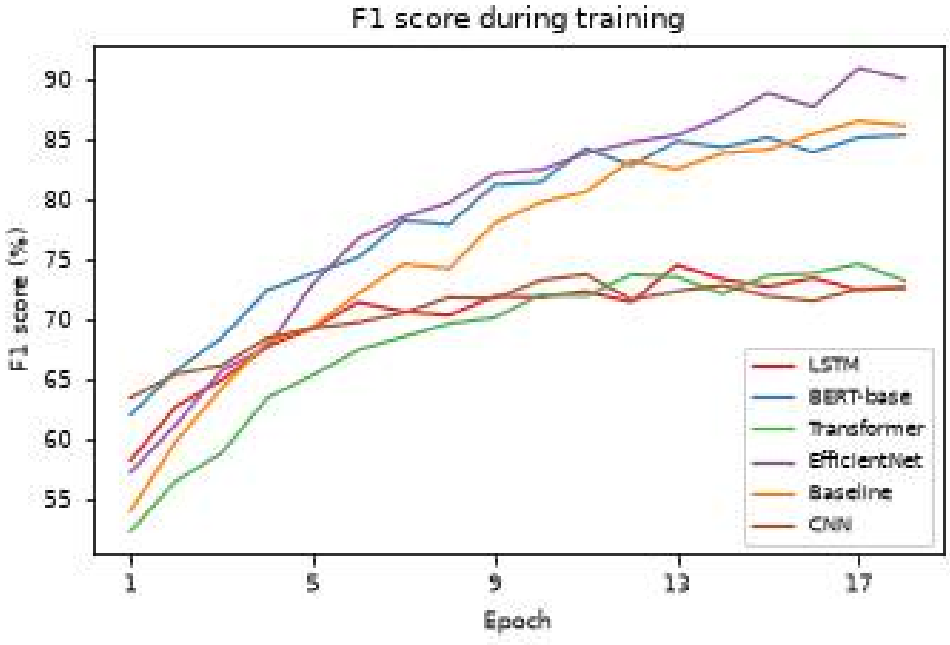


Figure 2: F1 score during training for the models in this experiment.

As shown in Figure 2, at epoch 13, Baseline is 9 points better than Transformer. BERT-base attains a F1 score of over 80% at epoch 11 (Figure 2). CNN attains a F1 score of about 70% at epoch 2. Baseline reaches 90.0% F1 score at epoch 18.

Figure 3 shows the accuracy during training.

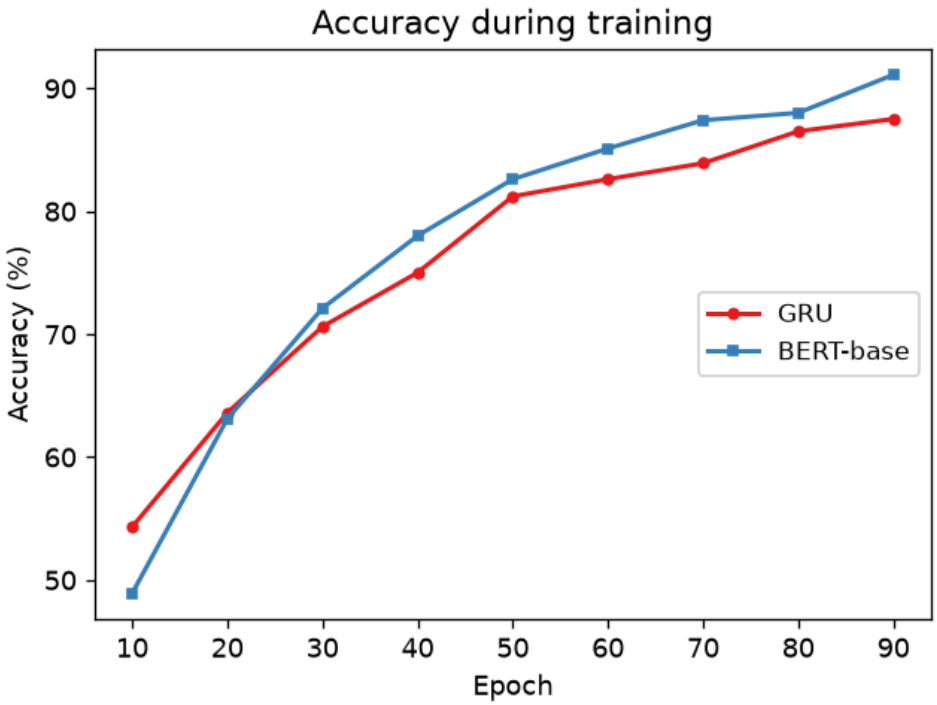


Figure 3: Accuracy during training for the models in this experiment.

At epoch 30, BERT-base achieves 62.8% accuracy. GRU reaches 54% accuracy at epoch 10. As shown in Figure 3, BERT-base reaches 88.0% accuracy at epoch 80.

3 Discussion

On STL-10, XGBoost obtains an accuracy of 64.4%. The differences between the models are discussed in the next section.